

main

243 Branches

161 Tags



Go to file

Go to file

Code



maxisbey and felixweinberger fix(server): read HTTP request bodies with a size limit and bound JSO...

7b781ed · 3 days ago

.changeset	fix(server): read HTTP request bodies with a size limit and bound JSO...	3 days ago
.github	docs: comprehensive GA pass — stable-release wording, released-spec l...	last month
common	fix(server): validate Content-Type by parsed media type instead of su...	last month
docs	fix(client/auth): propagate saveTokens errors after refresh (#2053)	last week
examples	docs: client log levels, elicitation completion notification; drive j...	last week
packages	fix(server): read HTTP request bodies with a size limit and bound JSO...	3 days ago
scripts	fix(scripts): stop the snippet sync walking node_modules and cloned f... (	2 weeks ago
test	docs: client log levels, elicitation completion notification; drive j...	last week
.git-blame-ignore-revs	Jerome/fix/dev oauth server required scopes change (#629)	last year
.gitignore	docs: rebuild the v2 documentation around a task-based page tree (#2397)	2 months ago
.npmrc	remove references to npm, clean up package manager usage in workspaces (	6 months ago
.prettierignore	Implement MCP 2026-07-28 (#2286)	2 months ago
.prettierrc.json	Implement MCP 2026-07-28 (#2286)	2 months ago
CLAUDE.md	Export Protocol and mergeCapabilities from client/server package roots (	last month
CODE_OF_CONDUCT....	run prettier (#995)	11 months ago
CONTRIBUTING.md	docs: comprehensive GA pass — stable-release wording, released-spec l...	last month
DEPENDENCY_POLIC...	docs: add ROADMAP, VERSIONING and DEPENDENCY_POLICY for the v2 line (#...	last week
LICENSE	chore: update licensing to Apache 2.0 for new contributions (#1377)	7 months ago
README.md	docs: comprehensive GA pass — stable-release wording, released-spec l...	last month
REVIEW.md	Implement MCP 2026-07-28 (#2286)	2 months ago
ROADMAP.md	docs: add ROADMAP, VERSIONING and DEPENDENCY_POLICY for the v2 line (#...	last week
SECURITY.md	Update SECURITY.md to use GitHub Security Advisories (#1555)	6 months ago
VERSIONING.md	docs: add ROADMAP, VERSIONING and DEPENDENCY_POLICY for the v2 line (#...	last week
lefthook-local.exempl...	v2: Add left-hook pre-push (#1003)	5 months ago
lefthook.yml	v2: Add left-hook pre-push (#1003)	5 months ago
package.json	docs: rebuild the v2 documentation around a task-based page tree (#2397)	2 months ago

pnpm-lock.yaml	docs: client log levels, elicitation completion notification; drive j...	last week
pnpm-workspace.yaml	fix(server): validate Content-Type by parsed media type instead of su...	last month
typedoc.config.mjs	docs: replace the typedoc site with VitePress (v2 + v1 reface) (#2395)	2 months ago
vitest.workspace.js	[v2] Monorepo, sdk-client, sdk-server, sdk-core package split (#...	8 months ago

[README](#) [Code of conduct](#) [Contributing](#) [License](#) [Security](#)

MCP TypeScript SDK

Important

This is the `main` branch — v2 of the SDK (`@modelcontextprotocol/server` , `@modelcontextprotocol/client`), implementing the [2026-07-28 MCP spec](#).

Have feedback? Please [open a v2 issue](#) — it is the most useful thing you can do for the SDK right now. The [v2 documentation](#) starts with a ten-minute server tutorial.

v2 is the **stable release line**, released alongside the 2026-07-28 spec. v1.x continues to receive bug fixes and security updates for at least 6 months after v2's release. v1 documentation: [ts.sdk.modelcontextprotocol.io](#) · v2: [/v2/](#) .

Warning

We're limiting pull requests to 1 per new contributor while v2 settles after the [2026-07-28 spec](#) release.

[Issues](#) are the most useful feedback right now — we'll reopen PRs as v2 stabilizes.

@modelcontextprotocol/server v2.0.0

@modelcontextprotocol/client v2.0.0

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► Table of Contents

Overview

The Model Context Protocol (MCP) allows applications to provide context for LLMs in a standardized way, separating the concerns of providing context from the actual LLM interaction.

This repository contains the TypeScript SDK implementation of the MCP specification. It runs on **Node.js**, **Bun**, and **Deno**, and ships:

- MCP **server** libraries (tools/resources/prompts, Streamable HTTP, stdio, auth helpers)
- MCP **client** libraries (transports, high-level helpers, OAuth helpers)
- Optional **middleware packages** for specific runtimes/frameworks (Express, Fastify, Hono, Node.js HTTP)
- Runnable **examples** (under [examples/](#))

Packages

This monorepo publishes split packages:

- `@modelcontextprotocol/server` : build MCP servers
- `@modelcontextprotocol/client` : build MCP clients

Tool and prompt schemas use [Standard Schema](#) — bring Zod v4, Valibot, ArkType, or any compatible library.

Middleware packages (optional)

The SDK also publishes small "middleware" packages under [packages/middleware/](#) that help you **wire MCP into a specific runtime or web framework**.

They are intentionally thin adapters: they should not introduce new MCP functionality or business logic. See [packages/middleware/README.md](#) for details.

- `@modelcontextprotocol/node` : Node.js Streamable HTTP transport wrapper for `IncomingMessage` / `ServerResponse`
- `@modelcontextprotocol/express` : Express helpers (app defaults + Host header validation)
- `@modelcontextprotocol/fastify` : Fastify helpers (app defaults + Host header validation)
- `@modelcontextprotocol/hono` : Hono helpers (app defaults + JSON body parsing hook + Host header validation)

Installation

Server

```
npm install @modelcontextprotocol/server
# or
bun add @modelcontextprotocol/server
# or
deno add npm:@modelcontextprotocol/server
```



Client

```
npm install @modelcontextprotocol/client
# or
bun add @modelcontextprotocol/client
# or
deno add npm:@modelcontextprotocol/client
```



Optional middleware packages

The SDK also publishes optional "middleware" packages that help you **wire MCP into a specific runtime or web framework** (for example Express, Fastify, Hono, or Node.js `http`).

These packages are intentionally thin adapters and should not introduce additional MCP features or business logic. See [packages/middleware/README.md](#) for details.

```
# Node.js HTTP (IncomingMessage/ServerResponse) Streamable HTTP transport:
npm install @modelcontextprotocol/node

# Express integration:
npm install @modelcontextprotocol/express express

# Fastify integration:
npm install @modelcontextprotocol/fastify fastify

# Hono integration:
npm install @modelcontextprotocol/hono hono
```



Getting Started

Here is what an MCP server looks like. This minimal example exposes a single `greet` tool over stdio:

```
import { McpServer } from '@modelcontextprotocol/server';
import { StdioServerTransport } from '@modelcontextprotocol/server/stdio';
import * as z from 'zod/v4';

const server = new McpServer({ name: 'greeting-server', version: '1.0.0' });

server.registerTool(
  'greet',
  {
    description: 'Greet someone by name',
    inputSchema: z.object({ name: z.string() })
  },
  async ({ name }) => ({
    content: [{ type: 'text', text: `Hello, ${name}!` }]
  })
);

async function main() {
  const transport = new StdioServerTransport();
  await server.connect(transport);
}

main();
```

Ready to build something real? Follow the step-by-step tutorials:

- [Build your first server](#) — a stdio weather-alert server, from `npm init` to a tool call
- [Build your first client](#) — connect to that server, list its tools, and call them

For runnable, end-to-end examples beyond the tutorials, see:

- [examples/README.md](#) — runnable, self-verifying client/server example pairs (one story per directory)

Documentation

- [Build a server](#) — your first MCP server, step by step
- [Build a client](#) — your first MCP client, step by step
- [Documentation site](#) — the full guides: tools, resources, prompts, serving over HTTP and stdio, clients, OAuth, and migration
- [Troubleshooting](#) — common errors and their fixes
- [API reference](#)
- [MCP documentation](#)
- [MCP specification](#)

Building docs locally

To work on the documentation site locally:

```
pnpm docs:api      # Generate the API reference markdown (output: docs/api/)
pnpm docs:dev      # Start the VitePress dev server for the V2 site
pnpm docs:build    # Build the V2 site (output: docs/.vitepress/dist/)
pnpm docs:multi    # Build the combined V1 + V2 site (output: tmp/docs-combined/)
```

The `docs:multi` script builds the V2 site from the current checkout, checks out the `v1.x` branch via a git worktree to build the V1 site, and produces a combined site with V1 docs at the root and V2 docs under `/v2/`.

v1 (legacy) documentation and fixes

If you are using the **v1** generation of the SDK, the **v1 API documentation** is available at <https://ts.sdk.modelcontextprotocol.io/>. The v1 source code and any v1-specific fixes live on the long-lived [v1.x branch](#). V2 API docs are at [/v2/](#).

Contributing

Issues and pull requests are welcome on GitHub at <https://github.com/modelcontextprotocol/typescript-sdk>.

Releases 159

 @modelcontextprotocol/fastify@2.0.0 Latest
last month

[+ 158 releases](#)

Contributors 182



[+ 168 contributors](#)

Languages

